

Darb – Your Journey, Your Way

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Abstract—The Darb Tourism Platform is a system designed to enhance tourism experiences in Saudi Arabia by combining curated tour planning with secure transportation services. The platform provides tourists with ready-made itineraries, limited customization options, real-time trip monitoring, and AI-driven personalized recommendations. To ensure safety and reliability, Darb verifies all drivers through official KSA government APIs and employs a Zero-Trust security model with continuous identity checks, encrypted data, and SOS emergency handling. The system supports tourists, drivers, and administrators through dedicated interfaces enabling tour browsing, driver verification, emergency management, and tour plan administration. Functional and non-functional requirements define the system’s behavior across key modules such as tour selection, AI recommendations, driver assignment, payment processing, and safety monitoring. Through its integrated approach and real-time features, Darb aims to provide a trustworthy, seamless, and culturally enriching tourism experience aligned with Saudi Arabia’s Vision 2030 goals.

I. INTRODUCTION

A. Purpose

The purpose of the Darb Tourism Platform is to provide a secure and intelligent system that enables tourists to browse ready-made tour plans across Saudi Arabia and book rides with verified drivers. The system supports limited adjustments to certain itineraries, such as changing one or two locations while maintaining the integrity and flow of the overall tour plan. This Software Requirements Specification (SRS) document guides developers, testers, designers, project managers, and stakeholders in understanding the system’s behaviour, constraints, and objectives.

B. Scope

The Darb Platform is a tourism system that connects tourists with pre-designed tour plans, AI-driven recommendations, and verified drivers. It enhances safety, convenience, and cultural engagement through structured itineraries, real-time tracking, and automated emergency support mechanisms.

1) System Goals:

- Provide curated, ready-made tour plans created by staff.
- Support limited modifications (1–2 location changes) where allowed.
- Ensure all drivers are verified through official KSA government APIs.
- Deliver AI-powered personalized recommendations.
- Enable real-time tracking, route monitoring, and SOS emergency alerts.

- Provide administrative tools for managing tours and driver verification.

2) System Benefits:

- Enhanced safety through verification and real-time tracking.
- Transparent pricing and predefined routes.
- Personalized experiences through AI recommendations.
- Reliable administrative oversight and secure trip monitoring.
- Improved tourism engagement aligned with Saudi Vision 2030.

3) System Boundaries: Included

- Ready-made tour plans.
- Limited modifications (1–2 stops).
- AI recommendations.
- Driver verification through KSA APIs.
- SOS functionality and trip monitoring.
- Real-time matching between tourists and drivers.
- Administrative management of tours and drivers.

Excluded

- Fully customizable or user-created tours.
- Free-form itinerary building.
- Offline booking or cash-only transactions.
- Physical vehicles or offline travel operations.

C. Definitions and Acronyms

Term / Acronym	Definition
Ready-made Tour	A curated, pre-designed itinerary containing multiple destinations.
Limited Modification	Ability to change 1–2 stops within allowed tour plans.
AI Engine	Module that generates personalized recommendations based on user persona and preferences.
Verified Driver	A driver validated through official KSA APIs for ID, license, and vehicle documents.
SOS	Emergency alert that sends real-time user location and details to staff.
Zero-Trust	Continuous identity, behaviour, and route verification model to ensure safety.
Persona	A user profile representing common tourism needs (solo traveler, family, friends’ group, etc.).
GPS/Map System	External service providing route tracking, deviation analysis, and real-time trip monitoring.

D. Competitor Analysis and Stakeholder Insights

1) *Competitor Analysis:* The Darb system integrates capabilities found in ride-hailing applications and tourism platforms, offering a unified solution focused on safety, cultural experience, and intelligent recommendations.

a) Main Competitors:

- **Uber / Uber Explore**
 - Strengths: Large network, convenient digital payments.
 - Weaknesses: No curated tours; limited safety monitoring.
- **Careem**
 - Strengths: Strong regional familiarity.
 - Weaknesses: Limited tourism-focused features.
- **GetYourGuide / Viator**
 - Strengths: Extensive catalog of tours.
 - Weaknesses: No personalized drivers; no continuous safety tracking.
- **Local Saudi Travel Agencies**
 - Strengths: Deep cultural knowledge.
 - Weaknesses: Lack of digital convenience and AI personalization.

b) Competitive Advantages of Darb:

- Ready-made tour plans tailored specifically to Saudi tourism.
- Limited modification option for controlled flexibility.
- Verified drivers supported by a Zero-Trust verification model.
- AI-driven personalized recommendations.
- Advanced safety features:
 - Live tracking.
 - SOS emergency alerts.
- Multi-language support and transparent pricing.

2) Stakeholder Insights:

a) Tourists Want:

- Verified drivers.
- Easy browsing of curated tour plans.
- Safety and live monitoring.
- Multi-language support.
- Authentic cultural experiences.

b) Drivers Want:

- Simple document upload.
- Fast verification process.
- Fair access to tour assignments.

c) Admins Want:

- Tools for creating and updating tour plans.
- Monitoring and response capabilities for emergencies.
- Full control over driver verification and approvals.

II. OVERALL DESCRIPTION

A. System Context

The system context defines how the Darb Tourism Platform interacts with all external entities. It illustrates the system

boundary and identifies key data exchanges between the platform and its actors.

a) External Entities:

- **Tourist/User:** Browses, customizes, and books tour plans.
- **Driver:** Uploads verification documents, accepts tour assignments, and updates trip status.
- **Admin/Staff:** Manages tour plans, verifies drivers, and handles emergency situations.
- **AI Engine:** Generates personalized tour recommendations.
- **Verification API:** Confirms driver identity and validates license and vehicle documents.
- **Payment Gateway:** Processes secure electronic payments for bookings.
- **GPS/Map System:** Tracks routes, monitors live trips, and triggers SOS alerts.

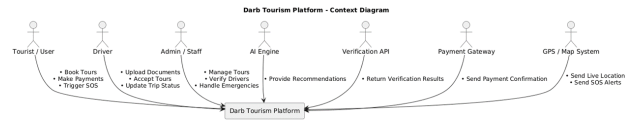


Fig. 1: System Context Diagram for Darb Tourism Platform

B. User Characteristics

The Darb Tourism Platform serves multiple user groups with diverse goals, behaviours, and technical abilities. Detailed personas were developed to represent these groups, guiding the design of features, user interactions, and safety requirements.

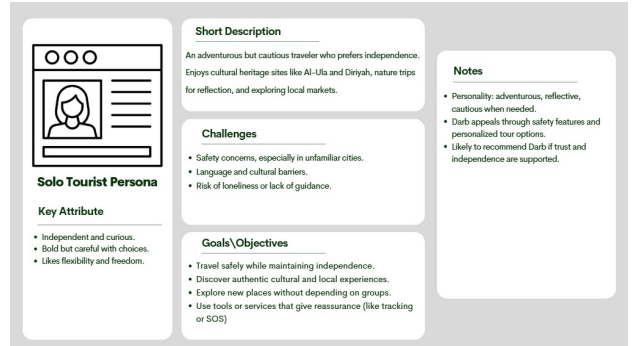


Fig. 2: Tourist Persona Example

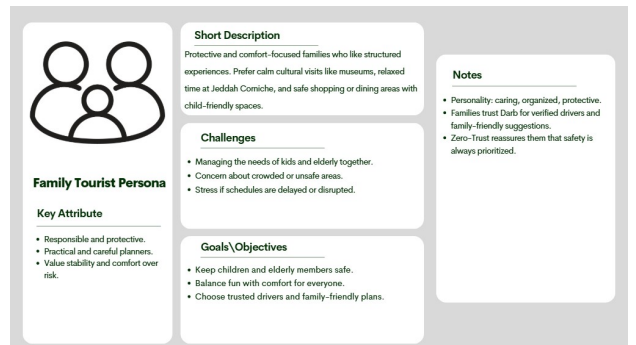


Fig. 3: Driver Persona Example

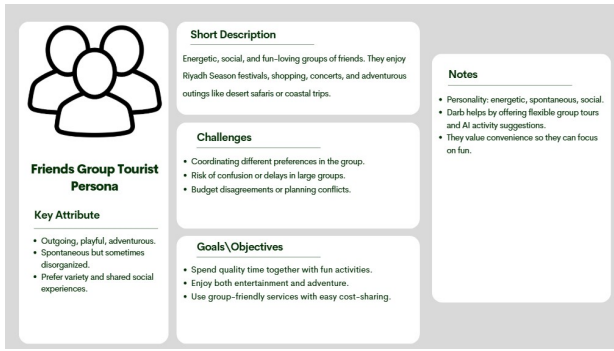


Fig. 4: Admin/Staff Persona Example

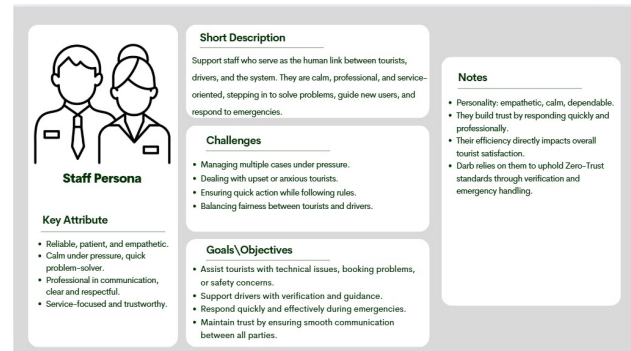


Fig. 8: Staff Persona Example

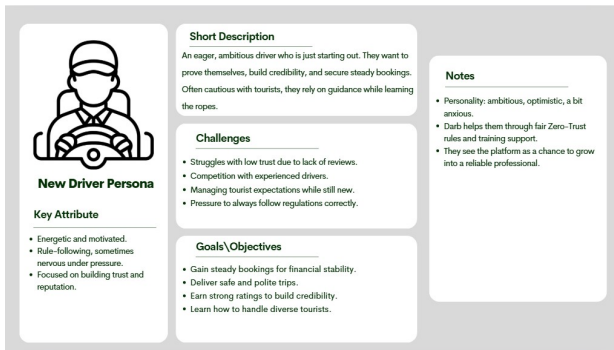


Fig. 5: New driver Persona Example

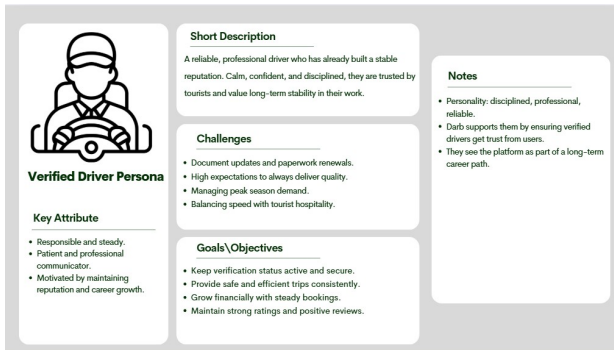


Fig. 6: Verified driver Persona Example

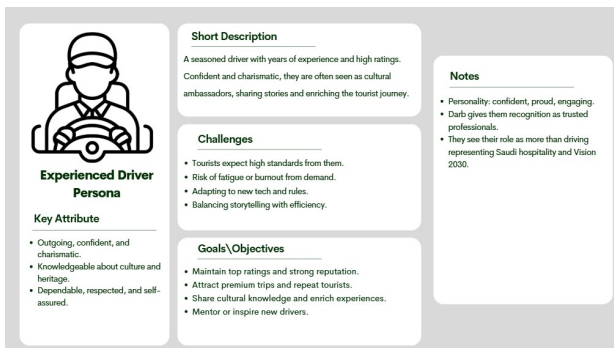


Fig. 7: Experienced driver Persona Example

C. Constraints

The Darb Tourism Platform is subject to several constraints that influence its design, implementation, and operational behaviour.

1) *Technical Constraints:* The platform must operate reliably on both iOS and Android environments. Core features—such as live tracking, route monitoring, SOS alerts, and driver verification—depend on continuous access to location services, network connectivity, and system notifications. Real-time responses, particularly for emergency alerts, must occur within five seconds. The backend must support high availability, scalability, and secure data logging to ensure stability during peak tourism seasons. The platform also relies on external services, including the AI recommendation engine, KSA government verification APIs, and online payment gateways.

2) *Security Constraints:* The system follows a Zero-Trust model that requires continuous identity and session verification for all user types. Any failed verification attempt must trigger automatic logout and notify staff. Sensitive data—including personal information, driver documents, and location history—must be encrypted both in transit and at rest. Administrative access requires multi-factor authentication, and only verified drivers are permitted to accept tour assignments.

3) *Regulatory Constraints:* The platform must comply with Saudi Arabian data privacy laws, tourism regulations, and official identity-verification procedures. All driver documents must be validated using approved KSA APIs. Emergency events must be logged in an auditable and tamper-proof manner.

4) *Operational Constraints:* SOS alerts must be processed and delivered to staff within a maximum of five seconds. Driver document verification must be completed within 48 hours of submission. Any updates made by staff, such as modifications to tour plans, must be reflected immediately across all user devices.

D. Assumptions and Dependencies

1) Assumptions:

- Users possess a compatible device and a stable internet connection.
- Tourists, drivers, and staff provide accurate information.

- Users grant the necessary permissions such as location access and notifications.
- Staff remain available to monitor and handle SOS alerts.
- Drivers upload valid and readable identification and vehicle documents.
- AI recommendations receive sufficient user data (persona, preferences).

2) Dependencies:

- KSA Government Verification API for driver document and identity validation.
- Payment gateways for secure transaction processing.
- GPS and mapping services for live tracking and SOS triggers.
- Notification services for delivering booking updates and alerts.
- Cloud backend for storage, logs, and real-time operations.
- AI engine for personalized tour recommendations.

III. FUNCTIONAL REQUIREMENTS

The following functional requirements define what the Darb Tourism Platform must provide in order to support tour browsing, AI-based recommendations, driver verification, emergency response, and administrative management.

ID	Name	Description
RQ1	View Plans	Tourists can view ready-made tour plans organized by category.
RQ2	Fetch Plans	System fetches and displays all available tour plans.
RQ3	Filter Plans	Filtering by location, duration, or budget with dynamic updates.
RQ4	Show Plan Details	Includes drivers, cost, and tour rating.
RQ5	Select Plan	User selects one plan for preview or booking.
RQ6	Store Selection	Selected plan stored in user's active session.
RQ7	Confirm Availability	System confirms availability and locks the plan.
RQ8	Adjust Tour details	User may adjust destinations or timing before booking.
RQ9	Validate Modifications	System validates changes for time, distance, and cost.
RQ10	Instant Update	All updates must reflect in under 5 seconds.
RQ11	Recalculate Itinerary	System recalculates duration and cost automatically.
RQ12	Recommend Personalized Tours	AI recommends up to 5 plans per user based on persona.
RQ13	Generate Recommendations	AI uses interests, budget, trip length, group size.
RQ14	Refresh Recommendations	AI refreshes after 3 completed bookings or profile edits.
RQ15	Upload Driver Documents	Driver uploads ID, license, vehicle docs.
RQ16	Verify via API	Documents verified using KSA Verification API.
RQ17	Complete Verification	Process must finish within 48 hours.
RQ18	Document Expiry Alerts	Alerts sent 7 days before document expiry.
RQ19	List Tours (Driver)	System lists tours matching driver region.
RQ20	Check Driver Status	System checks driver verification status.
RQ21	Confirm Assignment	Assignment confirmed and marked as "Booked."

ID	Name	Description
RQ22	Show Alternatives	Alternatives shown instantly if tour is taken.
RQ23	SOS Button	SOS visible on all active trip screens.
RQ24	Send SOS Alert	Sends location + trip details to staff within 5 seconds.
RQ25	Log Emergency	Logs time, coordinates, and device ID.
RQ26	Notify Support	Support staff notified instantly.
RQ27	Mark Resolved	Staff marks emergency as "resolved."
RQ28	Log Actions	Logs all support actions for auditing.
RQ29	Manage Plans	Staff can create, update, delete plans.
RQ30	Plan Details	Each plan includes name, duration, cost, category.
RQ31	Instant Sync	Updates reflect instantly to all users (5s).
RQ32	Continuous Verification	Continuous MFA verification for users and drivers.
RQ33	Route Deviation	Alert triggered if deviation > 500 meters.
RQ34	Auto Logout	Failed verification logs out user + alerts staff.

IV. NON-FUNCTIONAL REQUIREMENTS

The non-functional requirements of the Darb Tourism Platform are summarized in the table below. These requirements define quality attributes related to system performance, security, usability, reliability, scalability, and maintainability.

TABLE I: Non-Functional Requirements

ID	Category	Requirement	Description
NFR-1	Performance	Fast Loading	Pages and APIs must load in under 3 seconds.
NFR-2	Performance	Real-Time Alerts	SOS and location updates must be delivered within 5 seconds.
NFR-3	Security	Secure Login	Login must use multi-factor authentication (MFA).
NFR-4	Security	Data Protection	All sensitive data must be encrypted at rest and in transit.
NFR-5	Security	Document Safety	Driver documents stored securely and visible only to staff.
NFR-6	Reliability	Availability	System uptime must be at least 99.5%.
NFR-7	Usability	Easy to Use	Users must reach main features within two clicks.
NFR-8	Usability	Multilingual Support	Supports English and Arabic (RTL).
NFR-9	Scalability	Handle More Users	Must scale for peak tourism seasons.
NFR-10	Maintainability	Easy to Update	Modular design to simplify enhancements.

V. SYSTEM FEATURES / MODULES

This section describes the major features (modules) of the Darb Tourism Platform. Each feature includes its inputs, internal processing, and expected outputs.

A. Tour Plan Browsing & Customization

Inputs:

- User filters (location, duration, budget)
- Selected tour plan

- Optional modification request (change 1–2 places only in supported plan)

Processing:

- Fetch ready-made plans
- Apply filters and sorting
- Validate modification limits
- Recalculate cost and timing

Outputs:

- List of tour plans
- Detailed plan page
- Updated itinerary showing new cost & duration

B. AI Tour Recommendation

Inputs:

- User persona
- Interests, budget, trip duration, group size
- Trip history

Processing:

- AI Engine analyzes preferences
- Ranks suitable tour plans
- Refreshes recommendations on profile changes

Outputs:

- Up to 5 recommended tour plans
- AI relevance score/ranking

C. Driver Verification & Assignment

Inputs:

- Driver's ID, license, vehicle documents
- Region, vehicle type
- Tour selection by driver

Processing:

- Upload and validate documents
- Verify data via KSA API
- Check driver eligibility
- Assign driver to available tours

Outputs:

- Verification status (Pending / Verified / Rejected)
- List of available tours
- Booking assignment confirmation

D. SOS Alert & Emergency Handling

Inputs:

- Tourist SOS button press
- GPS route deviation
- Emergency logs from system

Processing:

- Capture user location & trip details
- Trigger SOS notification
- Create emergency event record
- Staff reviews and marks status

Outputs:

- Real-time SOS alert to staff
- Emergency log entry
- Resolved / unresolved emergency status

E. Tour Plan Management (Admin)

Inputs:

- New plan data (name, cost, duration, category)
- Edited plan details
- Delete request

Processing:

- Create or update tour plans
- Validate plan fields
- Update database instantly

Outputs:

- Updated list of plans visible to users
- Confirmation message to staff
- Audit log entry

F. Payment & Booking Processing

Inputs:

- User's selected plan
- Payment method (Apple Pay, Mada, etc.)
- Booking confirmation request

Processing:

- Validate payment
- Confirm availability
- Assign driver
- Generate booking record

Outputs:

- Payment confirmation
- Booking status (Booked / Failed)
- Receipt and trip info

G. Authentication & Continuous Verification

Inputs:

- User login credentials
- Session token
- Verification events (driver status, identity checks)

Processing:

- Authenticate users using MFA
- Issue and refresh session tokens
- Continuously verify identity during active sessions (Zero-Trust model)

Outputs:

- Active verified session
- Auto-logout on failed verification
- Staff alert if suspicious activity is detected

VI. EXTERNAL INTERFACES

This section describes all external interfaces that the Darb Tourism Platform interacts with, including user interfaces, hardware connections, software integrations, and communication standards.

A. User Characteristics

The platform includes three primary user interfaces:

Tourist Interface:

- Allows browsing and filtering of tour plans
- Displays plan details, prices, and available drivers
- Shows real-time trip tracking
- Includes an always-visible SOS button during active trips
- Supports Arabic and English

Driver Interface:

- Allows uploading ID, license, and vehicle documents
- Shows verification status
- Lists available tours filtered by region and vehicle type
- Allows accepting tour assignments
- Provides navigation and trip-tracking tools

Admin Dashboard (Web Interface):

- Allows creation, update, and deletion of tour plans
- Manages driver verification
- Displays SOS alerts in real time
- Provides emergency management tools and logs
- Includes authentication and access control for staff

B. Hardware Interfaces

The system interacts with the following device-level hardware features:

- **GPS Receiver** — Used for real-time route tracking and route deviation detection.
- **Sensors** — Used for location access during SOS events and live trips.
- **Network Connectivity (Wi-Fi / Mobile Data)** — Required for updating tour plans, sending notifications, and verifying documents.

No additional external physical hardware is required beyond standard smartphone sensors.

C. Software Interfaces

The system integrates with several external APIs and services:

KSA Verification API:

- Validates driver identity, license, and vehicle documents
- Returns verification status (Valid / Invalid / Expired)

Payment Gateway:

- Supports Apple Pay, Mada, credit cards
- Processes booking payments securely
- Returns payment confirmation or rejection

AI Recommendation Engine:

- Processes user data (persona, interests, history)
- Returns up to 5 recommended tour plans

GPS / Map Service:

- Provides real-time location tracking
- Calculates routes and durations
- Supports route deviation alerts ($\geq 500m$)

Notification Service:

- Sends booking updates, driver alerts, and SOS notifications
- Supports in-app and SMS delivery

Database Interface:

- Stores user profiles, tour plans, verification data, logs, and emergency records
- Allows secure read/write operations by backend components

D. Communication Interfaces

The system communicates using standard internet-based protocols:

- HTTPS (TLS 1.2+) for all API calls, login requests, payments, and verification
- RESTful API for backend services and mobile app communication
- WebSockets for real-time updates (SOS alerts, live location tracking)
- JSON format for request and response payloads
- OAuth2 / JWT for secure session management

VII. SYSTEM MODELS

A. Traceability Matrix

ID	Name	P	U1	U2	U3	U4	U5
RQ1	View Plans	M	✓				
RQ2	Fetch Plans	M	✓				
RQ3	Filter Plans	S	✓				
RQ4	Show Details	S	✓				
RQ5	Select Plan	M	✓				
RQ6	Store Plan	S	✓				
RQ7	Confirm Availability	M	✓				
RQ8	Adjust Tour	S	✓				
RQ9	Validate Mods	S	✓				
RQ10	Instant Update	S	✓			✓	
RQ11	Recalculate	S	✓				
RQ12	Recommend Tours	M	✓		✓		
RQ13	Generate Recs	S	✓	✓			
RQ14	Refresh Recs	C	✓	✓			
RQ15	Upload Docs	M			✓		
RQ16	Verify Docs	M			✓		
RQ17	Complete Verify	S			✓		
RQ18	Expiry Alerts	S			✓		
RQ19	List Tours	M			✓		
RQ20	Check Status	S			✓		
RQ21	Confirm Assign	S			✓		
RQ22	Alternatives	S			✓		
RQ23	SOS Button	M				✓	
RQ24	Send SOS	M				✓	
RQ25	Log Event	S				✓	
RQ26	Notify Staff	M				✓	
RQ27	Resolve Status	S				✓	
RQ28	Log Support	S				✓	
RQ29	Manage Plans	M					✓
RQ30	Plan Details	M					✓
RQ31	Update Users	M	✓				
RQ32	Continuous Ver.	M			✓		
RQ33	Route Deviation	M				✓	
RQ34	Auto Logout	M			✓	✓	

B. Use Case Diagrams

1) *UC1: Manage Tour Plans (Tourist)*: This use case describes how a tourist explores and interacts with the ready-made tour plans available on the Darb platform. The tourist can browse curated plans, apply optional filters, and view detailed information including cost, duration, and available verified drivers. If the selected plan supports limited customization,

the tourist may request to change 1–2 locations within the itinerary.

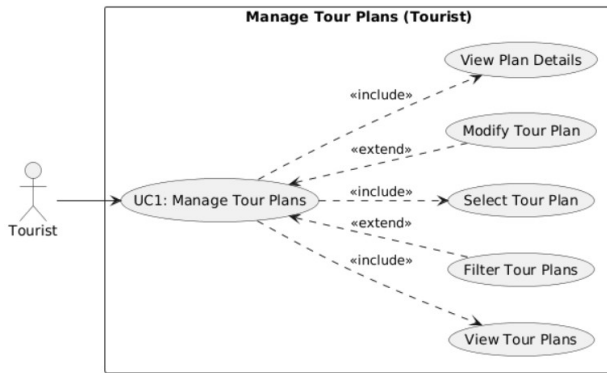


Fig. 9: Use Case Diagram for Manage Tour Plans (Tourist)

Field	Description
Primary Actor	Tourist
Preconditions	- Tourist is logged in. - Tour plans exist in the database.
Postconditions	- Selected plan stored in session. - Valid modifications updated in itinerary.
Main Flow	- Tourist opens tour plans. - System displays categorized plans. - Tourist applies filters. - System updates results instantly. - Tourist selects a plan. - System shows details. - Tourist modifies 1–2 places (if allowed). - System validates and recalculates itinerary. - Updated plan saved.
Extensions	4a: No results → system shows message. 7a: Modification exceeds limit → rejected. 8a: Invalid timing/distance → rejected.
Special Requirements	- Updates within 5 seconds. - Supports Arabic and English.
Constraints	- Only 1–2 modifications allowed. - Internet connection required.

2) *UC2: AI Tour Recommendation*: This use case explains how the Darb platform generates personalized tour recommendations for tourists. The AI engine analyzes multiple factors—including the user’s persona, interests, group size, and trip duration—to generate up to five recommended tour plans. The system automatically refreshes recommendations when the user updates their profile or completes several bookings.

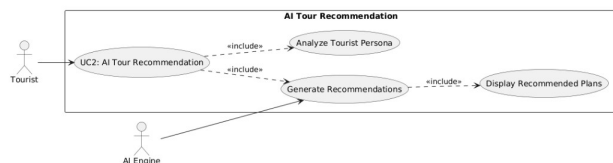


Fig. 10: Use Case Diagram for AI Tour Recommendation

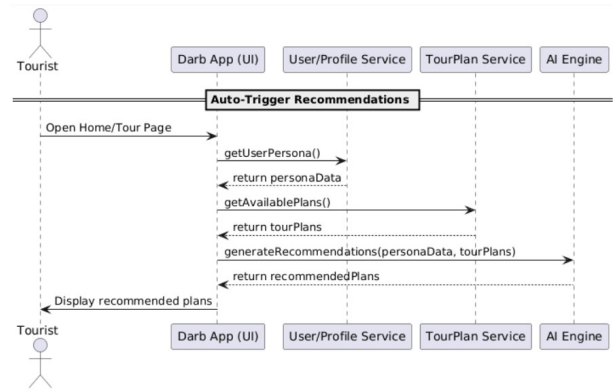


Fig. 11: Sequence Diagram for AI Tour Recommendation

Field	Description
Primary Actor	Tourist
Preconditions	- Tourist has persona profile. - AI engine is available.
Postconditions	Up to 5 recommendations displayed.
Main Flow	- Tourist opens recommendations. - System collects persona, interests, group size, duration, budget. - AI analyzes and ranks plans. - System displays top results.
Extensions	2a: Missing data → system requests profile update. 4a: No matches → system shows alternatives.
Special Requirements	Recommendations must be generated in real time.
Constraints	- AI uses ≥ 4 parameters. - Refreshes after profile update or 3 bookings.

3) *UC3: Driver Verification and Tour Selection*: This use case describes how drivers become verified and accept tours on the Darb platform. Drivers upload the required documents—including ID, driving license, and vehicle details—which the system validates through the official KSA Verification API. After verification is approved, the driver can view tours that match their region and vehicle type. When the driver selects a tour, the system performs the following steps:

- Checks whether the driver is currently verified.
- Confirms the availability of the selected tour.
- Assigns the tour and updates its status to “Booked.”

If the selected tour is already taken, the system instantly offers alternative available options.

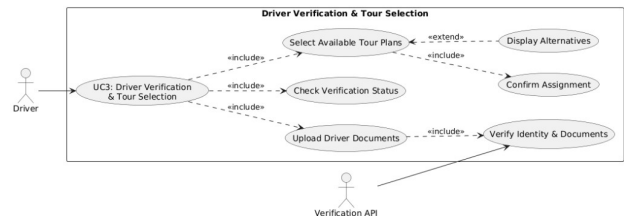


Fig. 12: Use Case Diagram for Driver Verification and Tour Selection

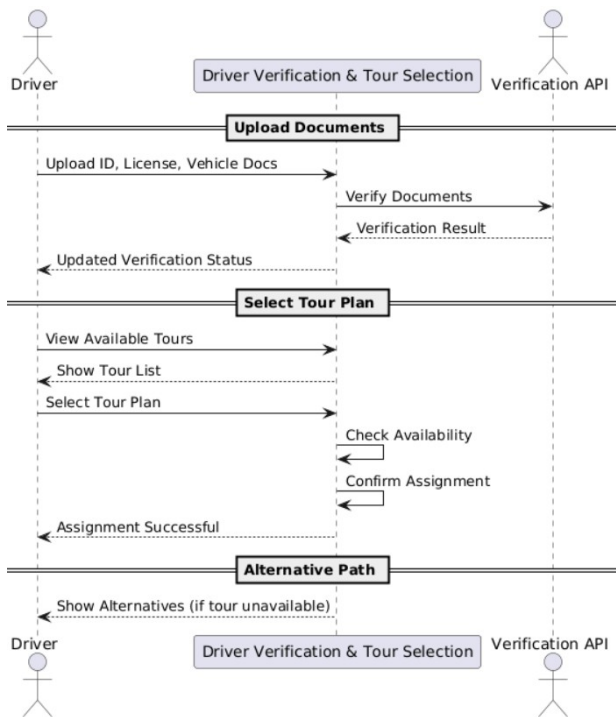


Fig. 13: Sequence Diagram for Driver Verification and Tour Selection

Field	Description
Primary Actor	Driver
Preconditions	- Driver has an account. - Documents ready for upload.
Postconditions	- Driver becomes Verified/Rejected. - Tour may be assigned.
Main Flow	1. Driver uploads ID, license, vehicle docs. 2. System sends to KSA API. 3. API returns status. 4. Verified driver sees available tours. 5. Driver selects a tour. 6. System validates status & availability. 7. System assigns tour and marks "Booked."
Extensions	• 3a: Rejected → system shows reason. • 5a: Tour taken → system shows alternatives. • 6a: Unverified driver blocked.
Special Requirements	• Verification completes within 48 hours. • Expiry alerts sent 7 days before.
Constraints	• Only verified drivers may accept tours.

4) *UC4: Emergency Alert and Response*: This use case describes how the Darb platform handles emergency situations during an active trip. The tourist may trigger the SOS button, which is always visible on the trip screen. Once the SOS button is pressed, the system performs the following actions:

- Captures the tourist's current location and trip details.
- Sends an alert to the support staff within five seconds.
- Creates an emergency log containing the timestamp and coordinates.

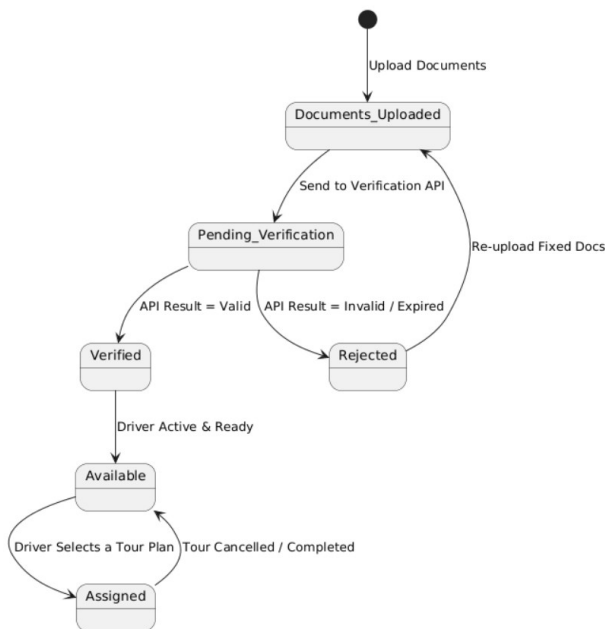


Fig. 14: State Diagram for Driver Verification and Tour Selection

Support staff review the alert, contact the tourist, and take the necessary actions to resolve the situation. After the incident is handled, staff mark the status as "Resolved," and all actions are recorded for auditing purposes.

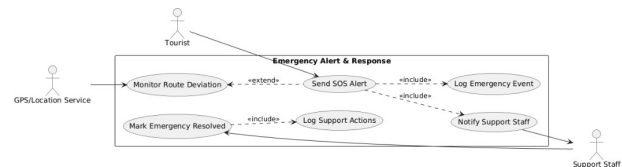


Fig. 15: Use Case Diagram for Emergency Alert and Response

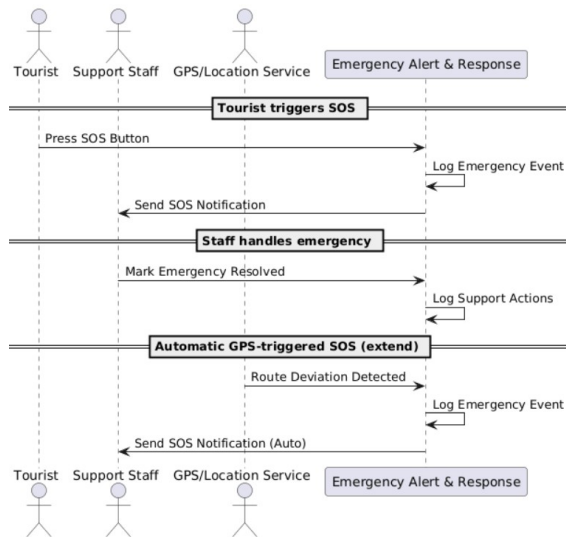


Fig. 16: Sequence Diagram for Emergency Alert and Response

Field	Description
Primary Actor	Tourist
Preconditions	- Tourist is on an active trip. - SOS button visible.
Postconditions	- Emergency logged. - Staff sets status to Resolved/Unresolved.
Main Flow	- Tourist presses SOS. - System captures location & trip details. - System sends alert to staff within 5 seconds. - Staff reviews alert. - Staff contacts tourist. - Staff marks emergency "Resolved." - System logs all actions.
Extensions	1a: No connection → system resends when network returns. 5a: Tourist unresponsive → escalate.
Special Requirements	- Real-time response (< 5 sec). - Accurate GPS tracking.
Constraints	SOS must appear on every active trip screen.

5) *UC5: Manage Tour Plans (Staff)*: This use case describes how administrative staff manage and maintain tour plans within the Darb platform. Staff members can create new plans, update existing ones, or delete outdated tours. Each tour plan must include essential attributes such as name, cost, duration, and category. Any modifications made by staff must be reflected to all users within five seconds. The system ensures:

- Input validation for all required fields.
- Instant synchronization of updates with all user interfaces.
- Logging of staff actions for accountability and auditing.

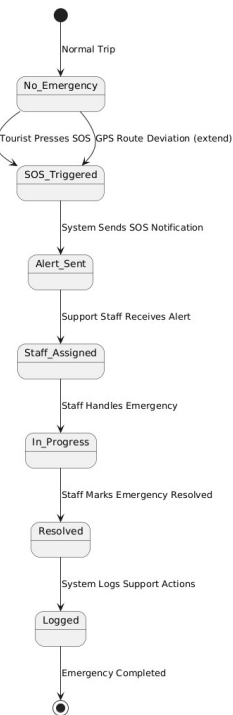


Fig. 17: State Diagram for Emergency Alert and Response

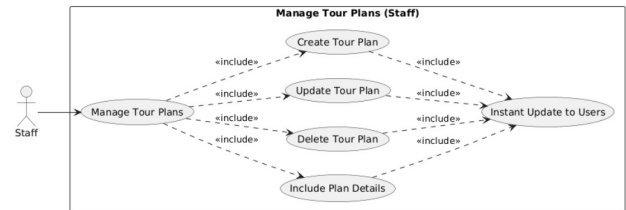


Fig. 18: Use Case Diagram for Manage Tour Plans (Staff)

Field	Description
Primary Actor	Staff / Admin
Preconditions	- Staff authenticated with MFA. - Tour plan data exists.
Postconditions	- Plans created, updated, or deleted. - Changes reflected to users.
Main Flow	- Staff opens dashboard. - Staff selects Create/Edit/Delete. - Staff enters required fields. - System validates input. - System saves changes. - Users see updates within 5 seconds.
Extensions	4a: Missing fields → error message. 5a: Database failure → retry/log error.
Special Requirements	- Instant updates (< 5 seconds). - Logging for accountability.
Constraints	Only authorized staff may modify plans.

C. Architectural Design

The Darb platform adopts a layered architectural design to ensure clear separation of concerns, improve maintainability, and support scalability. Each layer is responsible for a specific set of functions, allowing independent development, testing, and future enhancement.

Layer	Components	Responsibilities
1. Presentation Layer	Tourist App, Driver App, Admin Dashboard	Provides interfaces for interaction with the system.
2. Application Layer	Tour Management Service, AI Recommender, SOS Handler	Implements business logic, AI personalization, and safety processes.
3. Service Layer	Payment Gateway, Verification API, GPS Service	Handles external communication and APIs.
4. Data Layer	User DB, Driver DB, Tour Plans DB, Logs DB	Stores and manages persistent data.

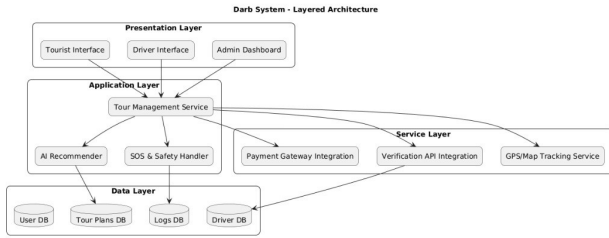


Fig. 19: Layered Architectural Design of the Darb Platform

D. Design Patterns

The following design patterns are used in the Darb Tourism Platform to improve flexibility, scalability, and maintainability.

1) *Strategy Pattern — Payment System*: The Strategy Pattern is used to support multiple payment methods within the platform. Different users may prefer various payment options such as credit cards, Apple Pay, or bank transfers.

a) *Reason for Use*: Different users prefer various payment methods (e.g., Credit Card, Apple Pay, Bank Transfer). The Strategy Pattern enables Darb to support multiple payment algorithms interchangeably.

b) *Classes Involved*:

- **PaymentStrategy** (interface)
- **CreditCardPayment**, **ApplePayPayment**, **BankTransferPayment**
- **PaymentContext**

Strategy Pattern - Darb Payment System

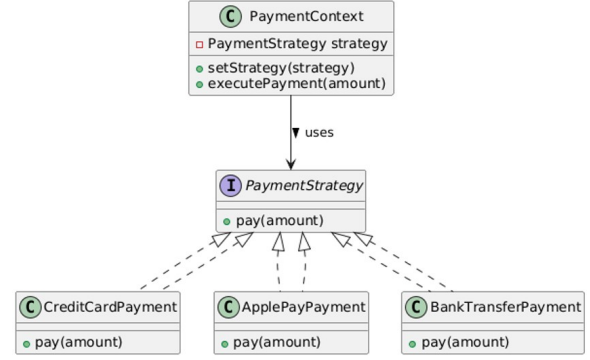


Fig. 20: Strategy Pattern Implementation for Payment Methods

2) *Observer Pattern — SOS Notification System*: The Observer Pattern is applied to manage the SOS alert mechanism within the Darb platform.

a) *Reason for Use*: When an SOS is triggered, multiple components must be notified including the Admin, Emergency Service, and Nearby Drivers. The Observer Pattern ensures all observers receive the update simultaneously.

b) *Classes Involved*:

- **SOSHandler** (subject)
- **AdminDashboard**, **EmergencyService**, **NearbyDriver** (observers)

Observer Pattern - SOS Notification System

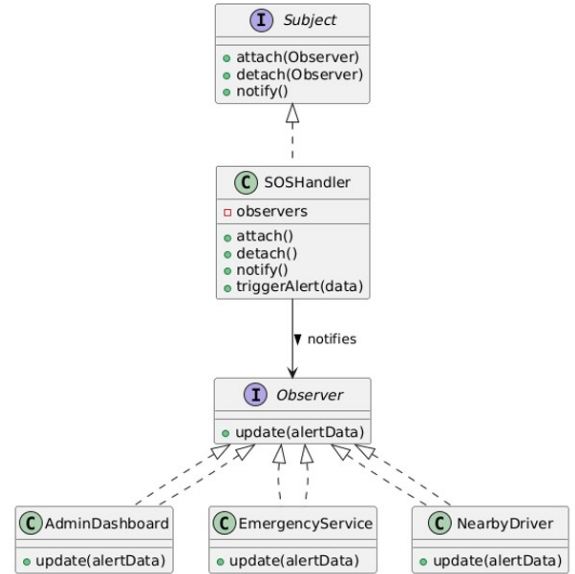


Fig. 21: Observer Pattern for SOS Notification System

3) *Adapter Pattern — AI Integration*: The Adapter Pattern is used to integrate the external AI engine with the Darb platform.

a) *Reason for Use*: The AI engine may return data in a format different from Darb's internal system. The Adapter Pattern allows the AI recommendations to be converted into a compatible structure without modifying existing classes.

b) *Classes Involved:*

- **AIRecommenderInterface** (target interface)
- **ExternalAIEngine** (adaptee)
- **AIAdapter** (adapter)

Adapter Pattern - AI Engine Integration

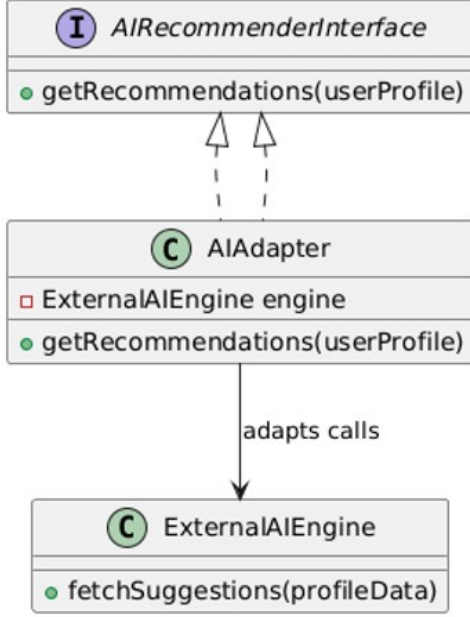


Fig. 22: Adapter Pattern for AI Integration

E. Class Diagram

The class diagram below represents the core domain concepts of the Darb Tourism Platform and illustrates how users, tours, bookings, and safety components interact. At the centre of the model is the abstract User class, from which Tourist, Driver, and Staff inherit, demonstrating that all three roles share common user attributes while performing different functions in the system. The diagram also highlights essential features of the platform such as tour planning, driver verification, personalized AI recommendations, and emergency handling.

Key relationships and responsibilities include:

- Tourists can create multiple **TourBookings**, receive **AI Recommendations**, and trigger **EmergencyEvents** during a trip.
- Drivers are assigned to bookings, maintain several **DriverDocuments**, and may be associated with emergency events.
- A **TourPlan** is composed of multiple ordered places through **TourPlanPlace**, allowing each tour to define a clear sequence of destinations.
- **EmergencyEvent** records SOS incidents and includes multiple **SupportActions** performed by **Staff**, ensuring corrective action is documented.
- **Places** represent real locations or points of interest that may appear in several tour plans.

Together, these classes define the core structure of Darb's domain and demonstrate how users, tours, documents, and safety processes connect to support a trustworthy and personalized tourism experience.

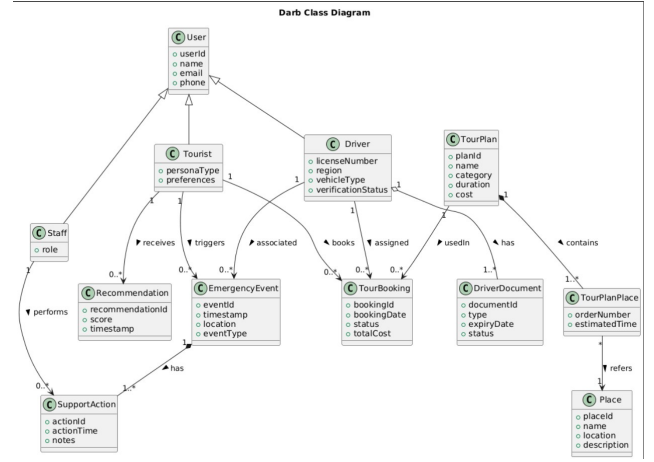


Fig. 23: Class Diagram of the Darb Tourism Platform

F. ER Diagram

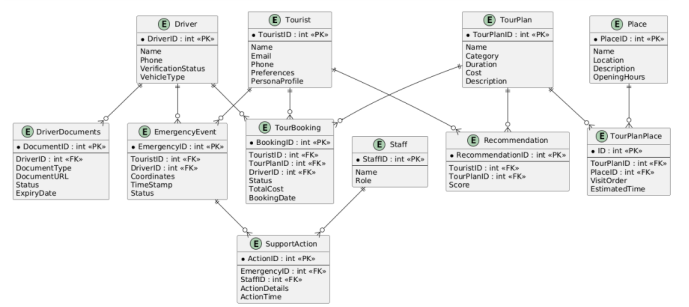


Fig. 24: ER diagram

Fig. 25: Entity-Relationship Diagram of the Darb Tourism Platform

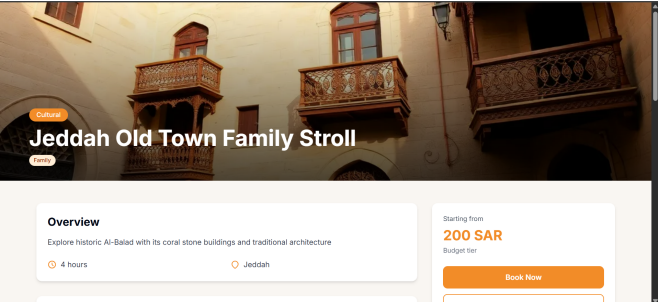
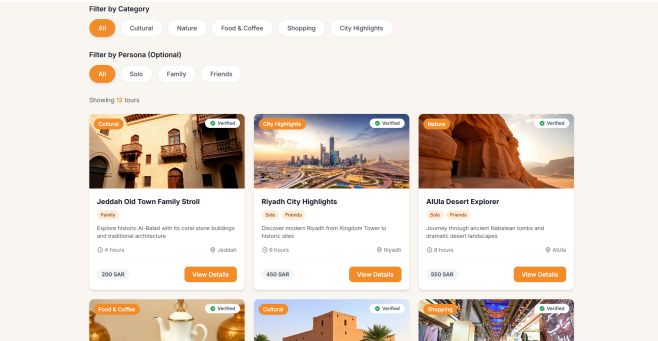
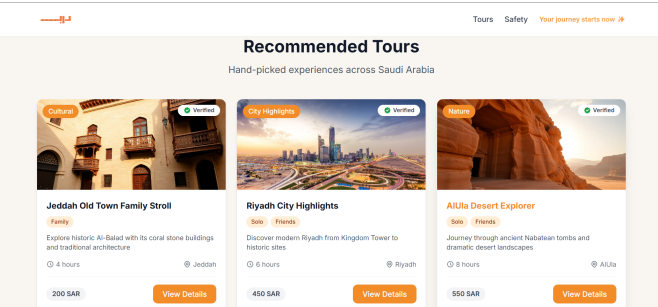
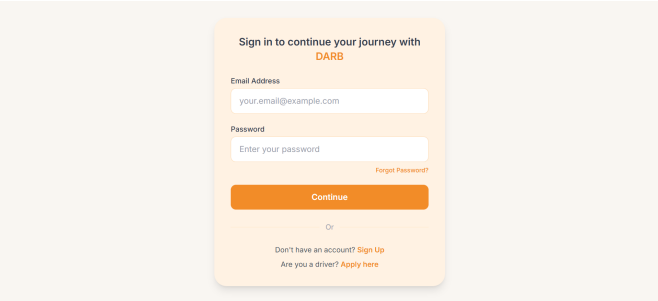
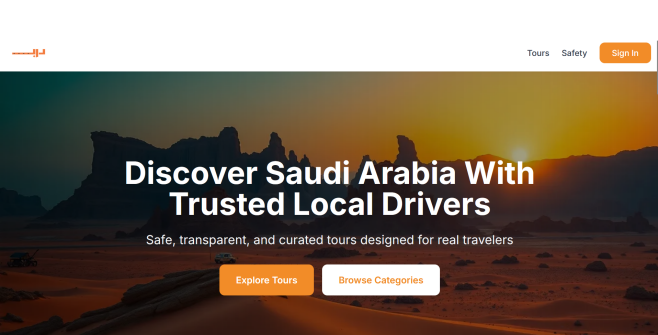
VIII. VALIDATION & ACCEPTANCE CRITERIA

This section describes how each functional requirement will be validated and the pass/fail conditions that must be met for the system to be accepted. The criteria are written in clear, measurable terms and directly map to the functional requirements.

Req ID	Requirement Summary	Validation Method	Acceptance Criteria (Pass/Fail)
RQ1	View tour plans	Functional test	System displays ≥ 5 plans per category.
RQ2	Fetch tour plans	API test	Plans load successfully from database.
RQ3	Filter tour plans	Functional test	Results update instantly based on filters.
RQ4	Display plan details	UI test	Driver info, rating, cost appear correctly.
RQ5	Select tour plan	Functional test	Only one plan can be selected at a time.
RQ6	Store selected plan	Session test	Selected plan remains saved during session.
RQ7	Confirm availability	Functional + DB test	Plan is locked for the user after confirmation.
RQ8	Adjust tour details	Functional test	User can modify 1–2 places only.
RQ9	Validate modifications	Logic test	System rejects invalid time/distance changes.
RQ10	Update within 5 sec	Performance test	Updates shown in ≤ 5 seconds.
RQ11	Recalculate itinerary	Functional test	New total cost/duration displays correctly.
RQ12	Recommendations list	Functional test	Up to 5 recommended plans appear.
RQ13	Use 4 parameters	Data test	Interest, budget, trip length, group size used.
RQ14	Refresh recommendations	Functional test	Updated after user profile/booking changes.
RQ15	Upload driver documents	UI test	System accepts images/PDFs.
RQ16	Verify via KSA API	Integration test	API returns Verified / Rejected.
RQ17	Complete in 48 hours	Time test	Verification done within 48 hours.
RQ18	Expiry alerts	Notification test	Alert sent 7 days before expiry.
RQ19	List tours for drivers	Functional test	Filtered by region + vehicle type.
RQ20	Check verification	Logic test	Unverified drivers blocked from accepting.
RQ21	Confirm assignment	DB + UI test	Status becomes “Booked.”
RQ22	Show alternatives	Functional test	Shows alternative tours instantly.
RQ23	SOS button visible	UI test	SOS button visible on trip screen.
RQ24	Send SOS alert	Performance test	Staff receives alert within 5 seconds.
RQ25	Log emergency	DB test	Time + coordinates stored.
RQ26	Notify staff	Functional test	Staff get real-time notification.
RQ27	Mark resolved	UI test	Staff can mark emergency as resolved.
RQ28	Log staff actions	DB test	All actions + timestamps recorded.
RQ29	Manage plans	Admin test	Admin can add/edit/delete plans.
RQ30	Required fields	Data test	Name, cost, duration, category required.
RQ31	Instant update	Performance test	Data updates in ≤ 5 seconds.
RQ32	Continuous verification	Security test	Failed verification logs out user.
RQ33	Route deviation alert	GPS test	Alert if deviation $> 500\text{m}$.
RQ34	Auto logout	Security test	User logged out automatically.

IX. UI/UX

The UI/UX design of the Darb platform emphasizes clarity, simplicity, and accessibility for all user roles. Tourists can easily browse and book curated tour plans, drivers can manage verification and assignments, and staff can monitor system activity efficiently. The design follows consistent layouts, recognizable icons, and minimal interaction steps to support quick decision-making and enhance overall usability.



Itinerary

1

Al-Balad Historic District

2

Naseef House

3

Traditional Souq

4

Old Jeddah Gates

5

Heritage Museum

6

Local Coffee Shop

Starting from

200 SAR

Budget tier

Book Now

Modify Plan

Safety Features

Verified Driver

Live Tracking

SOS Support

Zero-Trust Security

1234

Payment

Payment Details

Your Price

200 SAR

Total

200 SAR

Payment will be processed securely

Back

Continue

Verified Drivers

Ahmed Al-Rashid

Male

Toyota Land Cruiser

★★★★★ 4.9 (234)

✓

Muhammed Hassan

Male

Mercedes Sprinter

★★★★★ 4.8 (187)

✓

Khalid Ibrahim

Male

GMC Yukon

★★★★★ 4.7 (156)

✓

Laila Al-Harbi

Female

Lexus RX

★★★★★ 4.9 (210)

✓

Sara Al-Qahitani

Female

Hyundai Sonata

★★★★★ 4.8 (175)

✓

Starting from

200 SAR

Budget tier

Book Now

Modify Plan

Confirmation

✓

Booking Confirmed!

Your tour has been successfully booked

Live tracking will be available on tour day

View SOS Support

Return Home

verified Drivers

Ahmed Al-Rashid

Male

Toyota Land Cruiser

★★★★★ 4.9 (234)

✓

Muhammed Hassan

Male

Mercedes Sprinter

★★★★★ 4.8 (187)

✓

Khalid Ibrahim

Male

GMC Yukon

★★★★★ 4.7 (156)

✓

Laila Al-Harbi

Female

Lexus RX

★★★★★ 4.9 (210)

✓

Sara Al-Qahitani

Female

Hyundai Sonata

★★★★★ 4.8 (175)

✓

Starting from

200 SAR

Budget tier

Book Now

Modify Plan

preview-0m19311v-tourism-local-safety.deploypad.app says

Please select a driver before booking

OK

Sign in to continue your journey with DARB

Email Address

your.email@example.com

Password

Enter your password

Forgot Password?

Continue

Or

Don't have an account? Sign Up

Are you a driver? [Apply here](#)

Become a Darb Driver

Join our network of verified local drivers



Multi-Step Verification

Our rigorous verification process ensures safety and quality for all travelers.

1

Identity Verification

2

Background Check

3

Vehicle Inspection

4

License Validation

5

Training Completion

Upload Documents

Submit required documents securely

Get Verified Badge

Earn your trusted driver status

Start Earning

Accept tours and grow your income

Ready to Join?

Start your application today and become part of Saudi Arabia's trusted tourism network

Apply Now

1234

Tour Review

Jeddah Old Town Family Stroll



Duration: 4 hours

City: Jeddah

Price: 200 SAR

Continue

1234

Date & Time

Date

12/24/2025

Time

11:57 PM

Back

Continue

Zero-Trust Safety Model

Your safety is our top priority



Layered Security Approach

Our Zero-Trust model means we verify everything, trust nothing. Every driver, every trip, every moment is monitored to ensure your complete safety.

Identity Verification

Real-time document verification

Behavior Monitoring

Real-time driver behavior analysis

Route Anomaly Detection

AI-powered route deviation alerts

SOS Support

24/7 emergency response system

Live Tracking

Real-time GPS tracking for all trips

Compliance Checks

Ongoing vetting and quality assurance

24/7 Support

Round-the-clock assistance

Verified Drivers

All drivers thoroughly vetted

Live Tracking

Track your trip in real-time

X. APPENDICES

A. Business Model

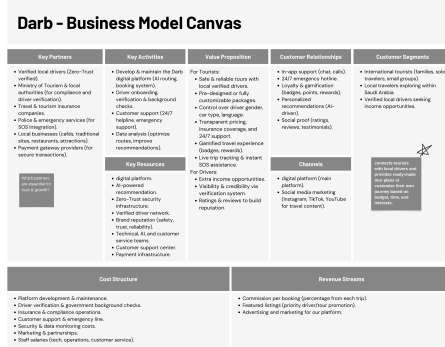


Fig. 26: Business Model

B. Gantt Chart

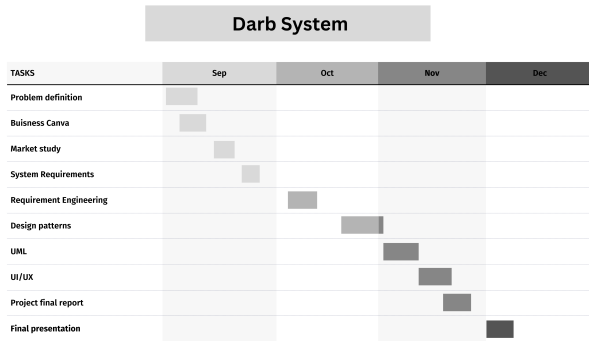


Fig. 27: Project Gantt Chart

C. References

- 1) Statista. (2024). *Tourism in Saudi Arabia – Market Insights & Data*. <https://www.statista.com/topics/8895/tourism-in-saudi-arabia/>
- 2) UNWTO (2023). *Tourism Market Trends in the Middle East*. United Nations World Tourism Organization. <https://www.unwto.org/tourism-data>
- 3) GetYourGuide. (2024). *Customer Reviews & Feedback on Tours*. <https://www.getyourguide.com/>
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D. Glossary

Term	Definition
Admin / Staff	System operators responsible for creating and updating tour plans, verifying driver documents, and handling SOS emergencies.
AI Engine	Component that analyzes user persona, preferences, and history to generate personalized tour recommendations.
Booking	A confirmed reservation made by a tourist for a specific tour plan and assigned driver.
Driver	A verified user who provides transportation services and must maintain valid ID and vehicle documents.
Driver Document	Official documents such as ID, driving license, and vehicle registration uploaded by drivers and verified through the KSA API.
Emergency Event (SOS Event)	A recorded incident triggered by a tourist, including location, time, and event details.
GPS / Map System	External service responsible for route tracking, trip monitoring, and detecting route deviations.
KSA Verification API	Official Saudi government API used to validate driver identity, license, and vehicle documentation.
Limited Modification	Feature allowing tourists to change 1–2 stops within certain tour plans without altering the tour structure.
Persona	A user profile that represents typical tourist behavior and preferences (e.g., solo traveler, family tourist).
Place	A point of interest or destination included in one or more tour plans.
Recommendation	AI-generated personalized tour suggestion for tourists.
Route Deviation	A situation where a vehicle deviates more than 500 meters from the planned route, triggering alerts or safety checks.
SOS Button	A safety feature that allows tourists to send emergency alerts with live location to staff.
Support Action	A logged action performed by staff to respond to and resolve an emergency event.
TourBooking	A record linking a tourist, tour plan, and driver, including booking details and status.
Tourist	A user who browses tour plans, receives recommendations, makes bookings, and can trigger SOS alerts.
Tour Plan	A curated itinerary created by staff that includes multiple destinations in a structured order.
Tour Plan Place	A component of a tour plan that assigns order and estimated time to each place in the itinerary.
User	A general entity representing any platform participant (Tourist, Driver, or Staff).
Verification Status	The state of a driver's document validation process (e.g., Pending, Verified, Rejected).
Zero-Trust Model	A security approach based on continuous identity verification, monitoring, and session validation.